

OpenStack RCA Copilot

Evidence-first, graph-first root cause investigation for OpenStack

Project Presentation

July 13, 2026

Table des matières (1/4)

- ❶ Problem: OpenStack failures are hard to debug
- ❷ Goal: turn logs into investigation evidence
- ❸ Current deployment topology
- ❹ Current implemented pipeline
- ❺ Evidence lineage and transformations
- ❻ Log ingestion: journald to MongoDB

Table des matières (2/4)

- ⑦ Raw evidence storage
- ⑧ Parsing: raw logs to structured events
- ⑨ Graph correlation: events to relationships
- ⑩ Graph safety controls
- ⑪ Incident detection: suspicious seed events
- ⑫ Incident construction: bounded subgraphs

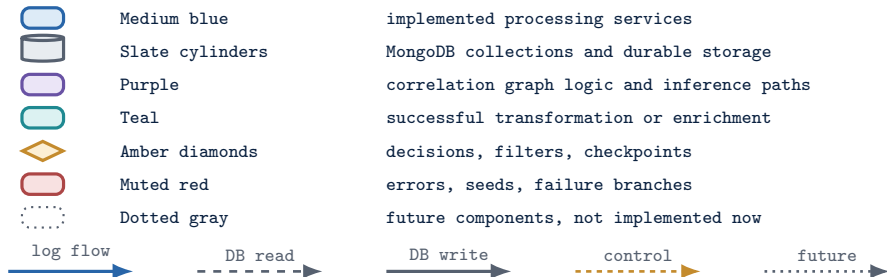
Table des matières (3/4)

- 13 Worker reliability
- 14 Enrichment: deterministic evidence package
- 15 MongoDB collection architecture
- 16 Failure containment
- 17 Current versus future boundary
- 18 Future AI/RAG layer

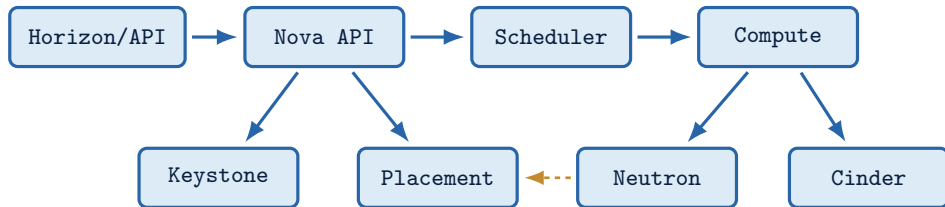
Table des matières (4/4)

- 19 Future local provider node
- 20 Future grounded AI workflow
- 21 Horizon dashboard and operator workflow
- 22 Operator investigation journey
- 23 Final architecture
- 24 Roadmap and next steps

Semantic Visual Language

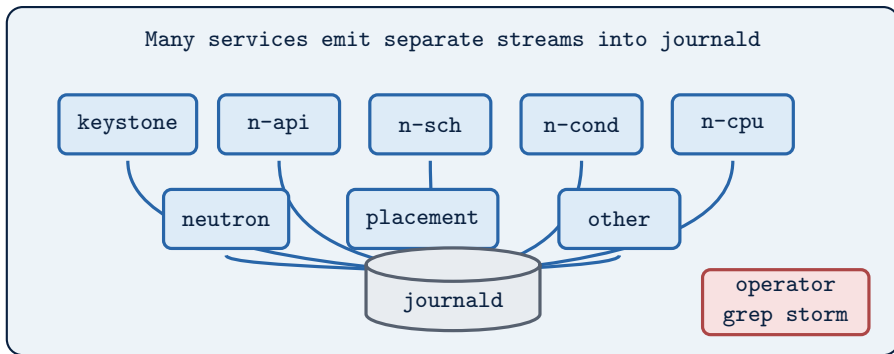


OpenStack Service Ecosystem

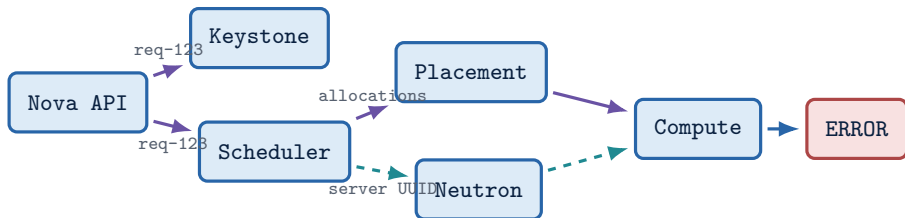


A single operation crosses independent services, logs, IDs, and clocks.

Distributed Log Volume

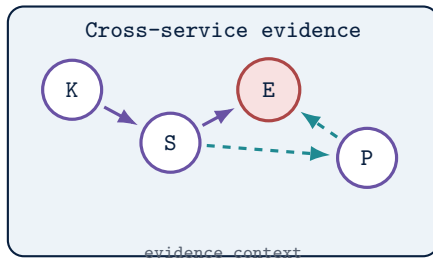
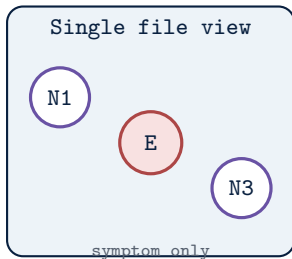


One Request Crosses Many Services

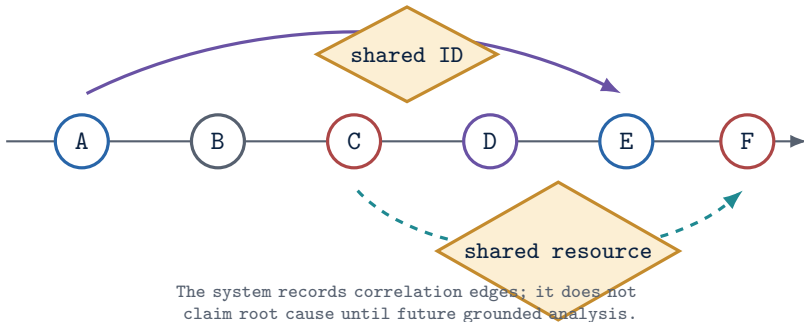


The visible failure may be far from the first relevant event.

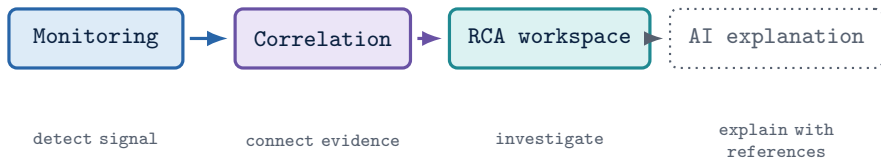
Why Isolated Log Inspection Fails



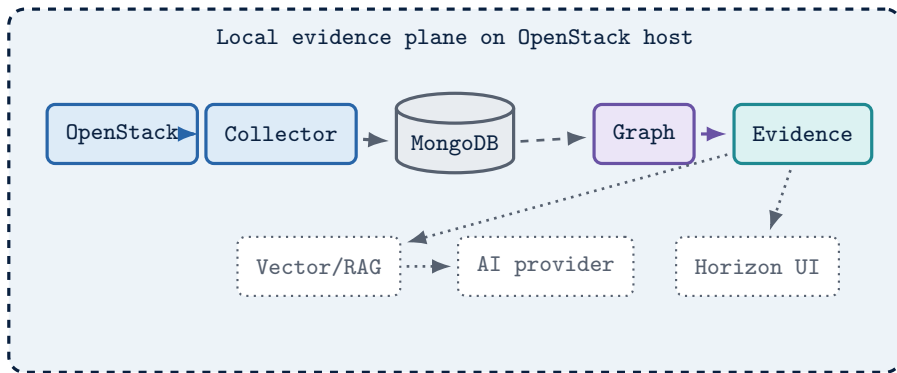
Chronology Is Not Causality



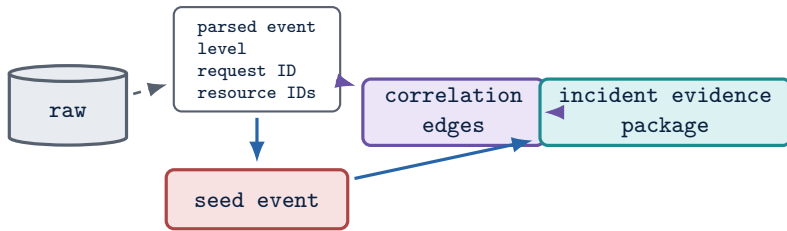
Monitoring vs Correlation vs RCA vs AI



Vision Architecture



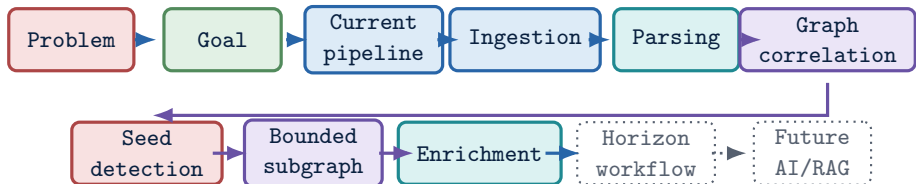
Evidence-First Design



AI is downstream of evidence, never a substitute for collected facts.

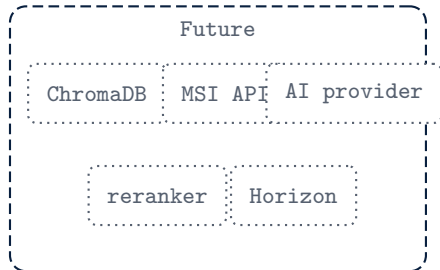
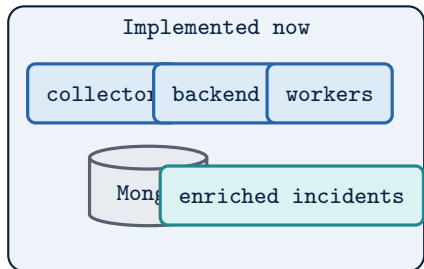
Project Storyline

Read the deck as one evidence pipeline

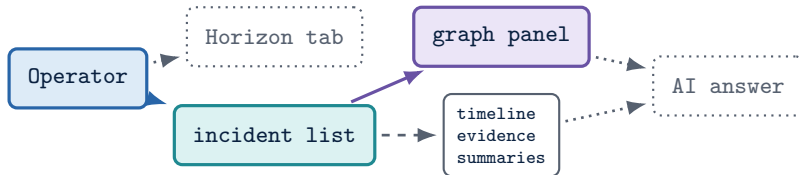


Each step shows the input, the transformation, the stored artifact, and the operator-facing result.

Implemented vs Future System

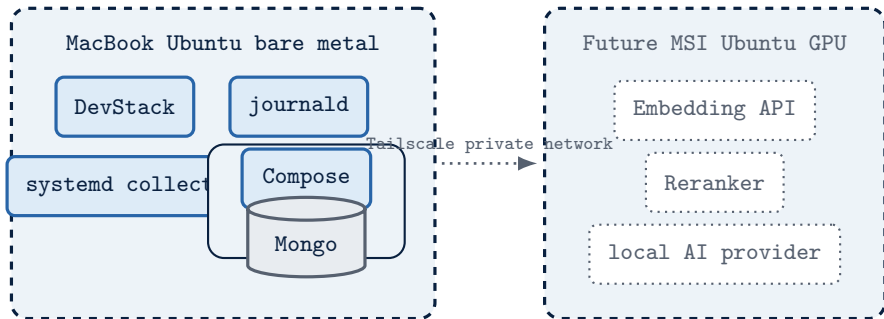


Operator Interaction Model

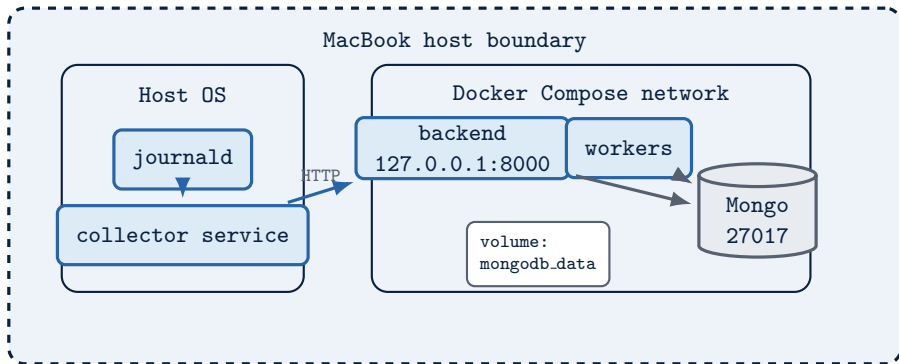


The browser never calls AI providers directly; backend mediates future inference.

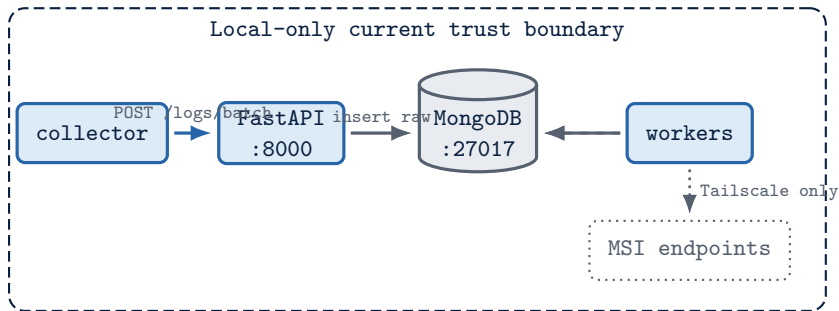
Two-Machine Physical Topology



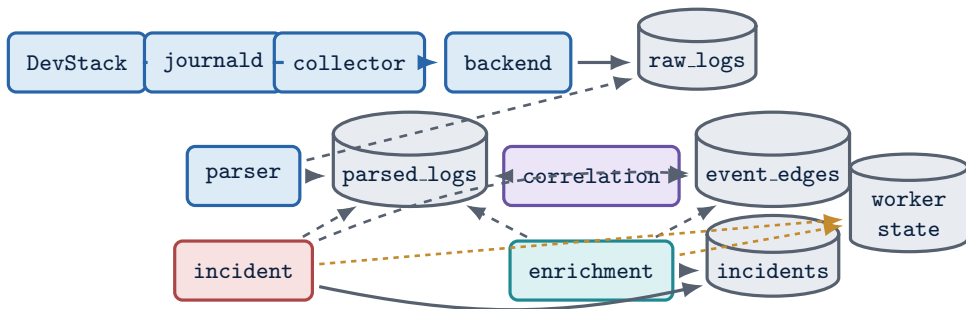
MacBook Host and Container Boundaries



Ports, Protocols, and Trust Boundaries



Runtime Dependency Graph

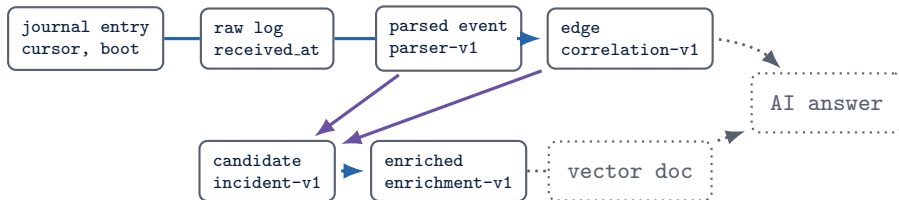


Future Runtime Extensions

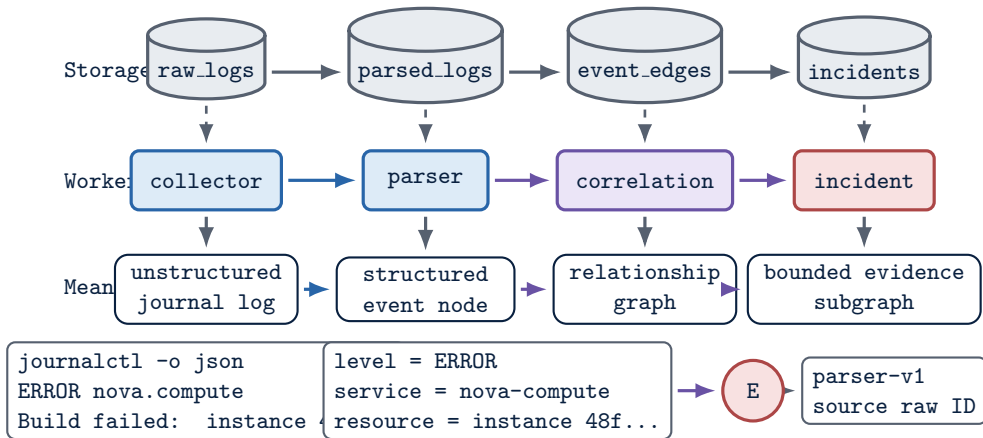


MongoDB remains source of truth; Chroma stores vectors and metadata only.

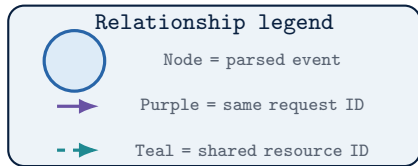
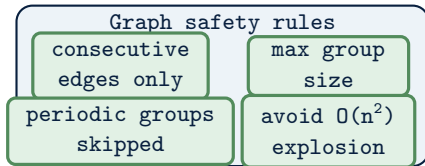
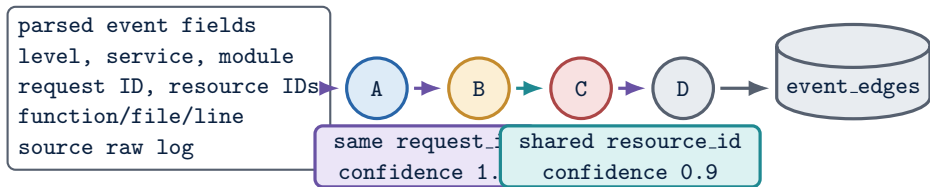
Data Lineage Map



Parsing and Graph Relationships



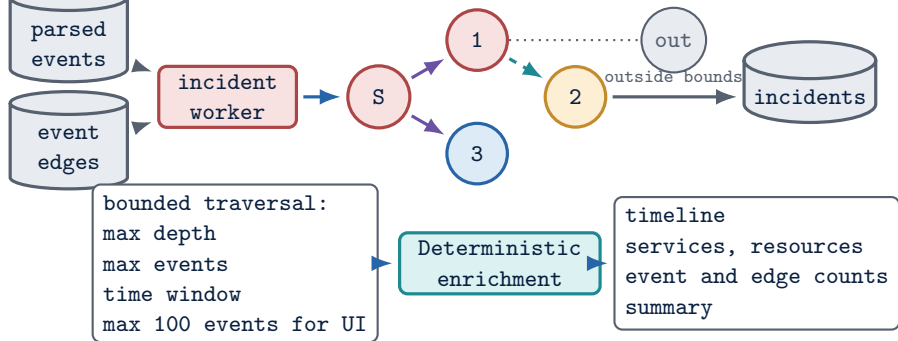
Correlation: Structured Events to Graph Edges



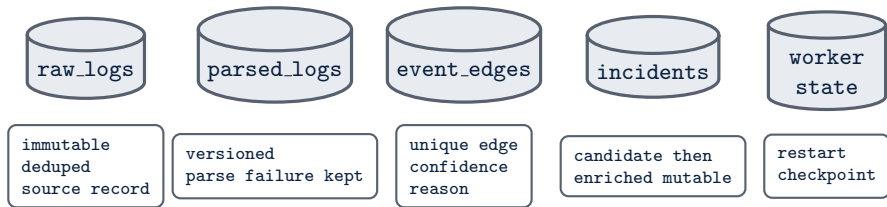
Edges are correlation, not causality

Incident Evidence Package

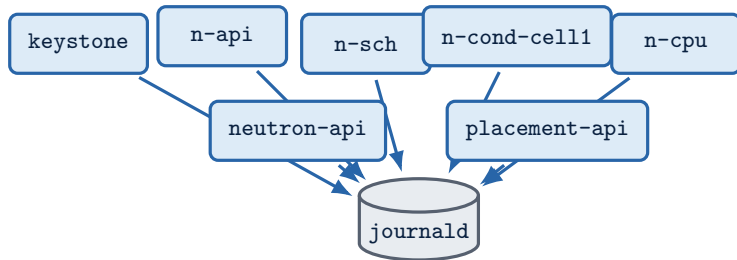
Seed event \ bounded traversal over event_edges → incident evidence subgraph



Artifact Responsibilities

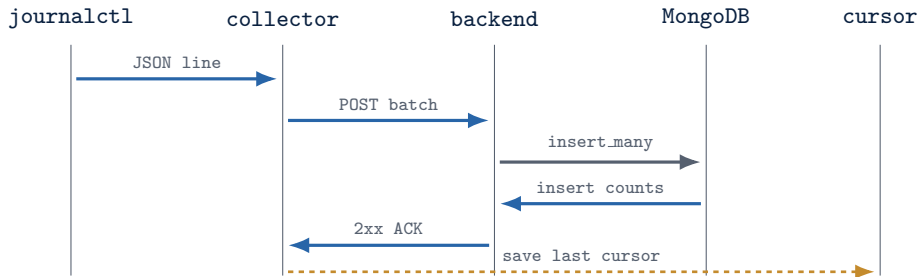


Monitored DevStack Units

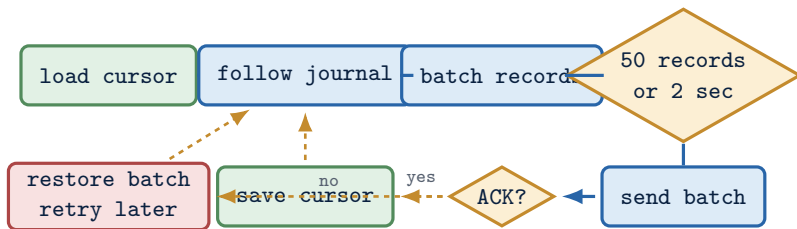


Default unit list from collector/config.py

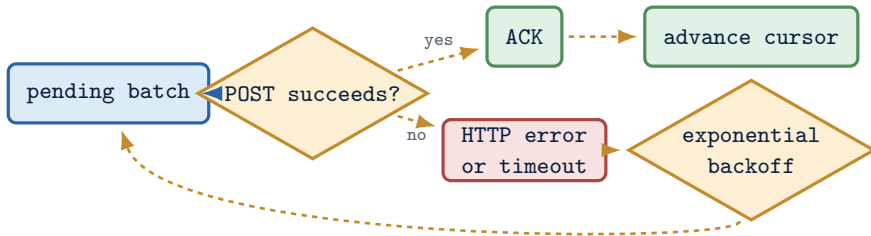
Collector Sequence: Cursor Advances After ACK



Collector State Machine

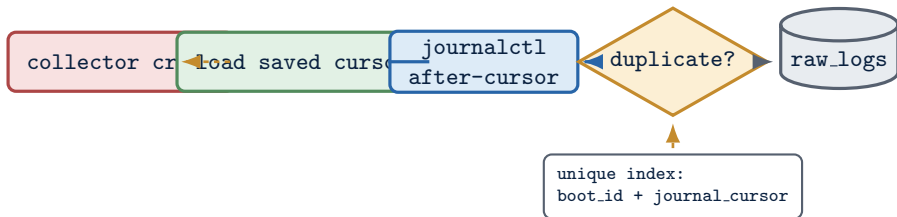


Backend Unavailable: Retry Branch



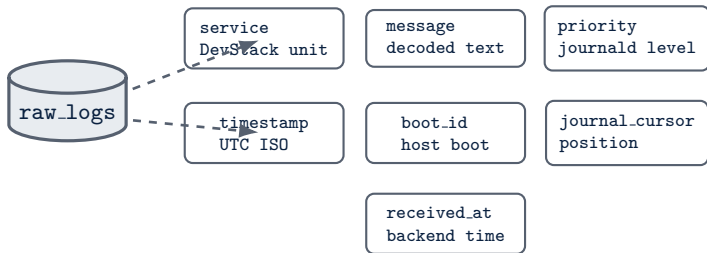
Failed batches are restored before sleeping; cursor is not advanced.

Crash Recovery and Duplicate Prevention

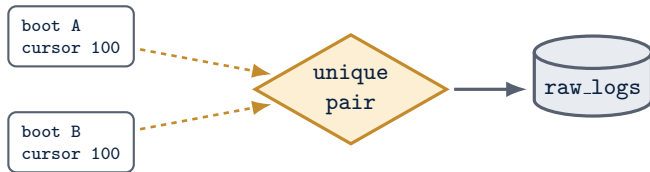


Replayed records are counted as duplicates instead of corrupting the raw collection.

raw_logs Schema as Evidence Envelope

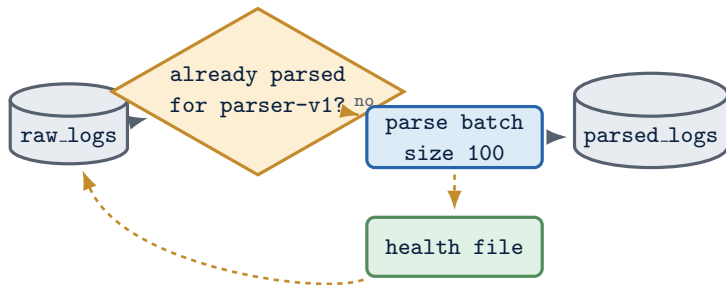


boot_id + journal_cursor Identity

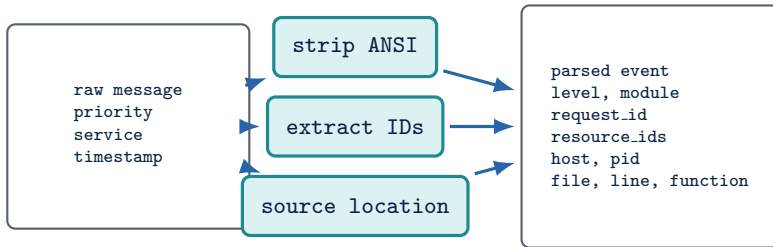


Cursor values are scoped by boot ID, so the pair is the stable journal identity.

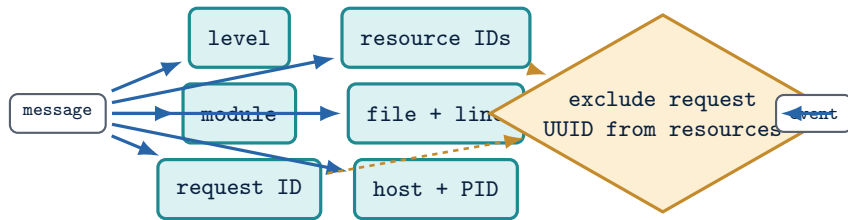
Parser Worker Lifecycle



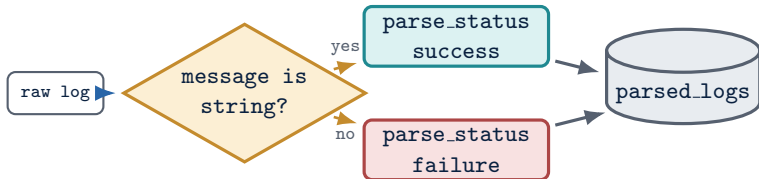
Raw Text to Structured Event



Extraction Graph

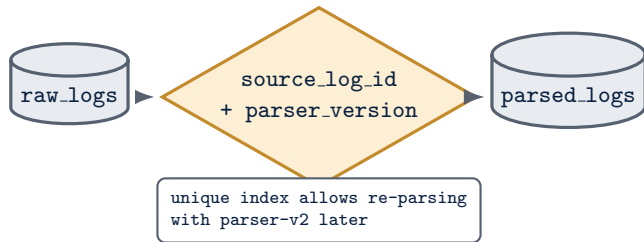


Parse Failure Preservation

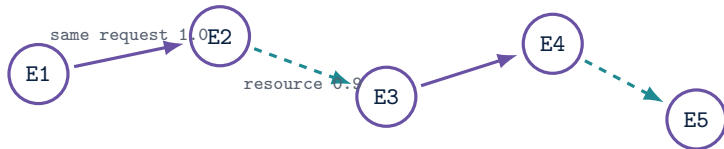


Bad records stay visible and auditable rather than disappearing.

Parser Idempotency and Versioning

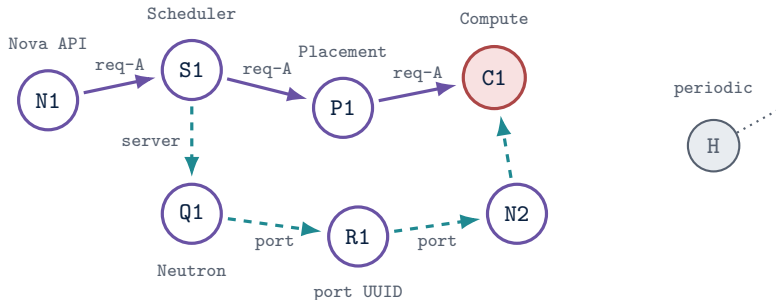


Event Nodes and Correlation Edges

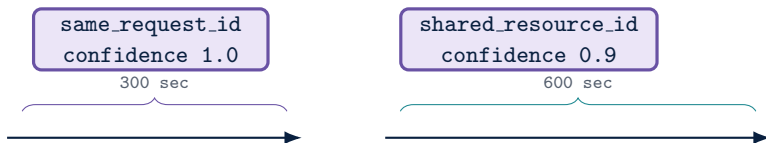


Edges are chronological: source event is earlier, target event is later.

Realistic Multi-Service Correlation Graph

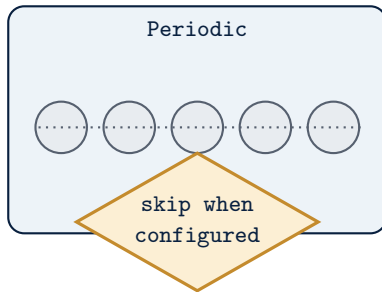
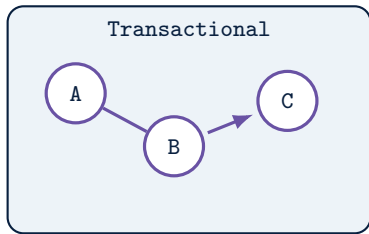


Correlation Rule Windows

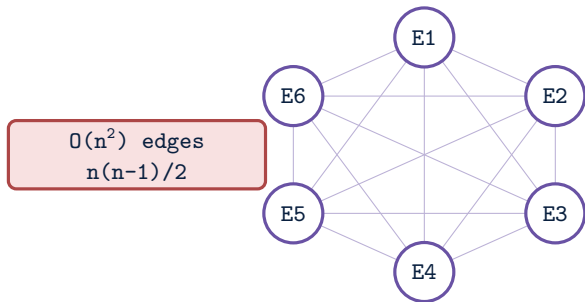


Rules fetch neighboring events inside each group window.

Transactional vs Periodic Groups



Before: Complete Pairwise Graph



Dense graphs hide signal and amplify repeated periodic IDs.

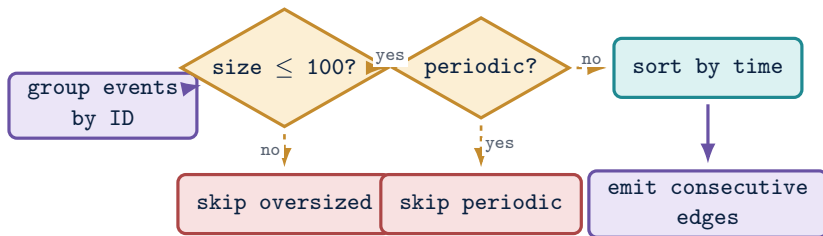
After: Consecutive-Only Chain



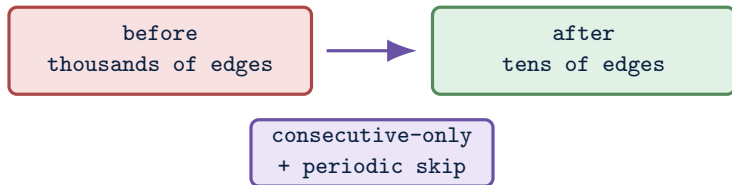
maximum $n-1$ edges

Events are sorted by timestamp and linked only to the next event in the group.

Explosion Fix Decision Flow

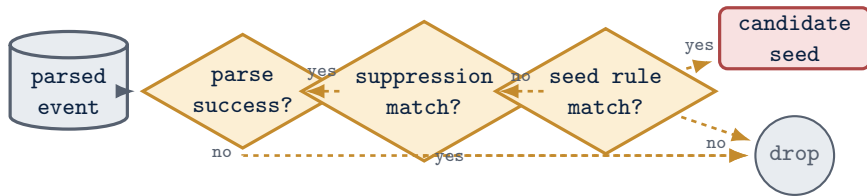


Observed Impact

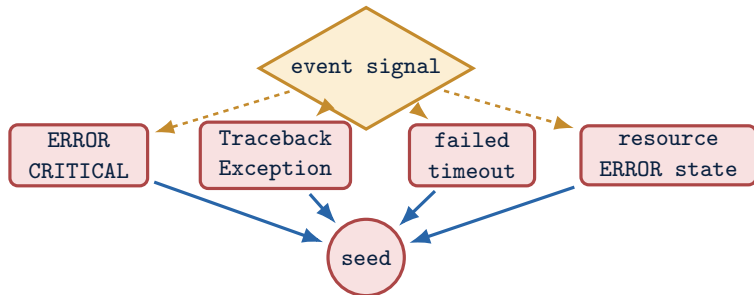


This is an engineering control against false correlation, not just a performance optimization.

Incident Seed Rule Engine

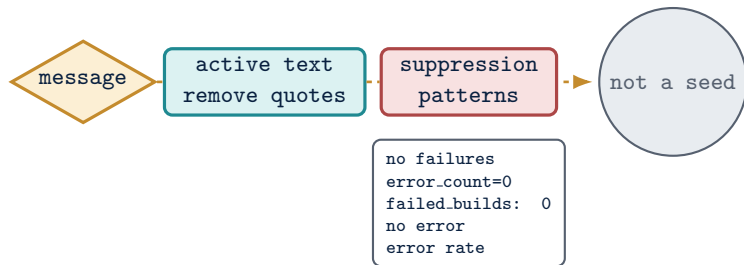


Accepted Seed Decision Tree

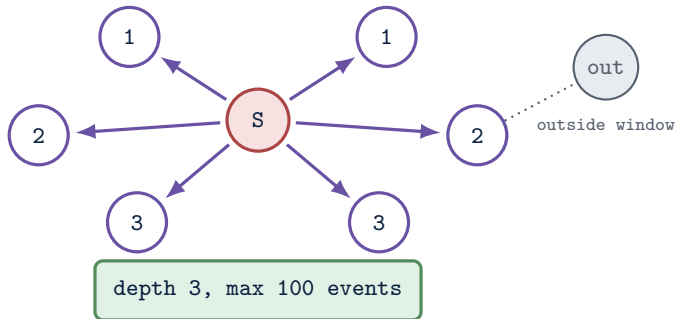


Severity is CRITICAL only when parsed level is CRITICAL; otherwise ERROR.

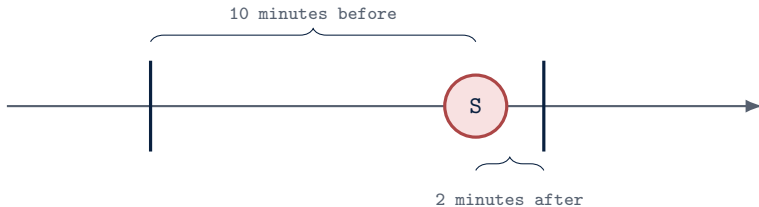
Rejected False Positives



Bounded Traversal Around Seed

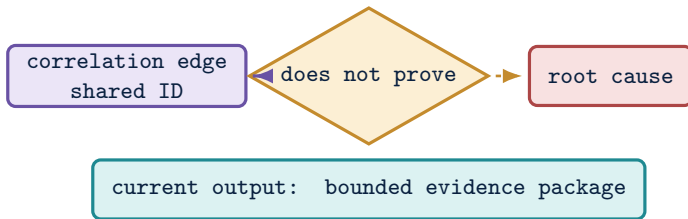


Incident Window

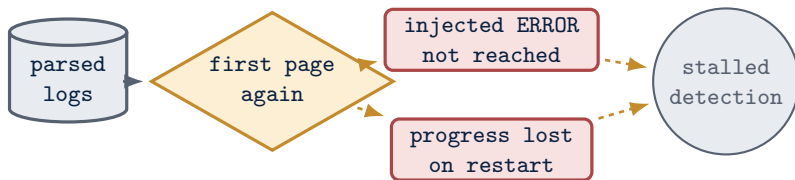


Only events inside the window can join the incident subgraph.

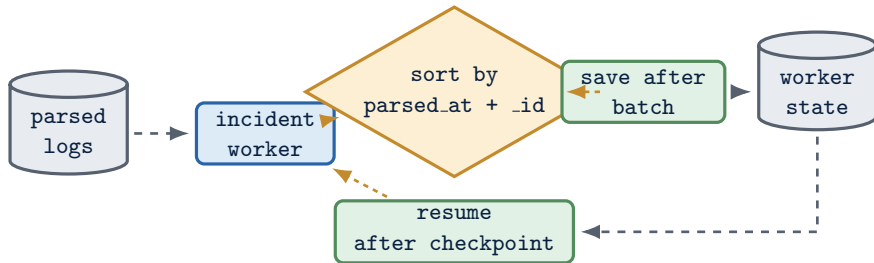
Correlation Is Not Causality



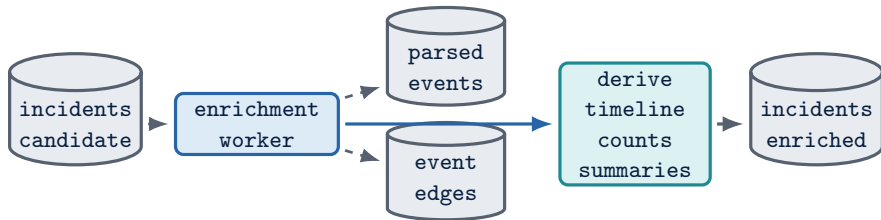
Before: Repeated First-Page Scanning



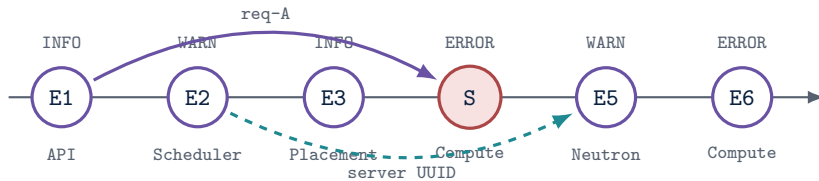
After: Persistent worker_state



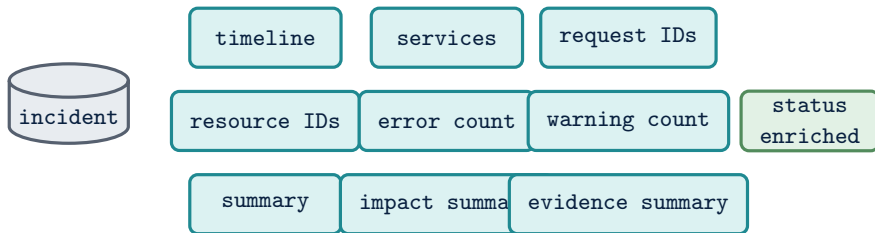
Enrichment Worker Architecture



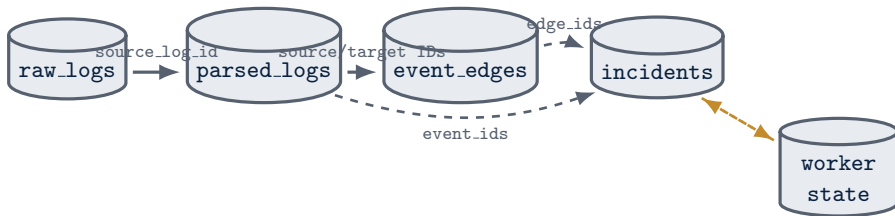
Enriched Incident Timeline



Enriched Incident Document



Collection Relationship Map



Indexes and Version Fields

<code>raw_logs</code> <code>uniq boot_id + cursor</code>	<code>parsed_logs</code> <code>uniq source + pars</code>	<code>event_edges</code> <code>uniq edge + reason +</code>	<code>incidents</code> <code>uniq seed + incident version</code>
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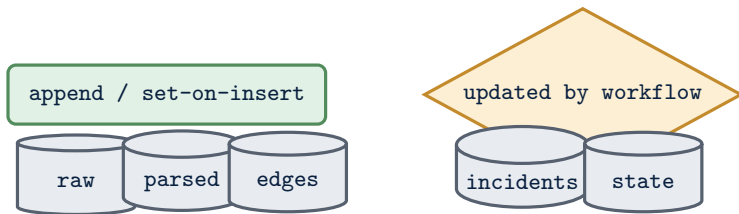
`parser-v1`

`correlation-v1`

`incident-v1`

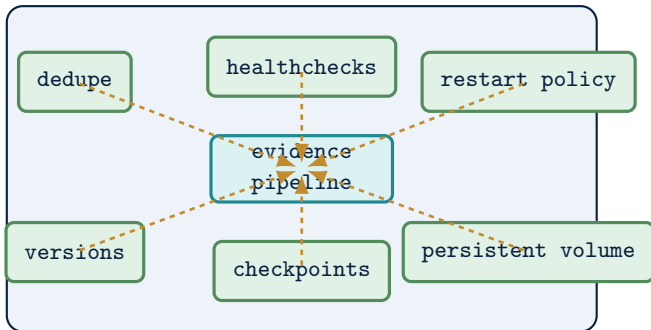
`enrichment-v1`

Immutable vs Mutable Collections

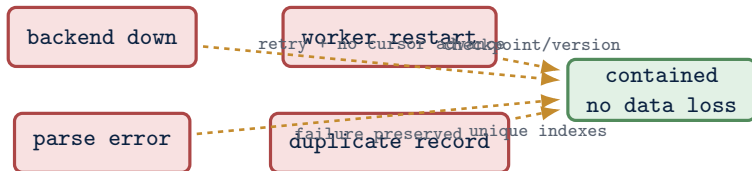


Raw evidence is preserved; incident documents mature from candidate to enriched.

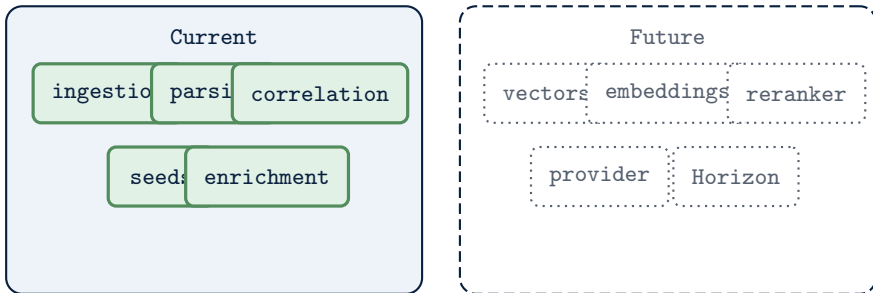
Reliability Shield



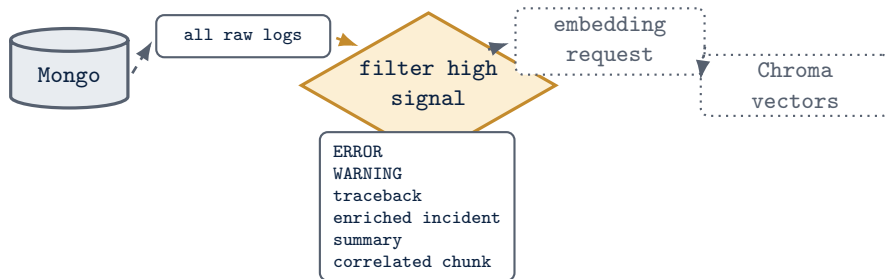
Failure Containment



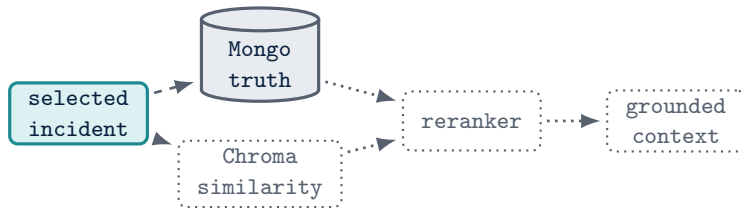
Current and Future Capability Boundary



Selective Vectorization Funnel

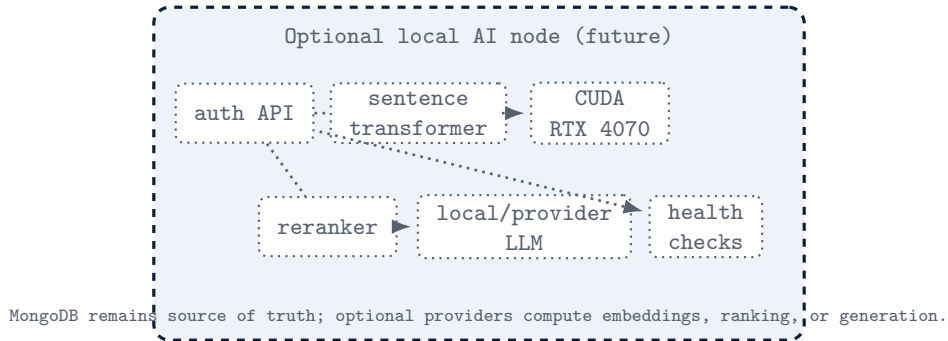


Retrieval Architecture



Retrieved vectors point back to MongoDB evidence IDs.

Future Local Provider Node



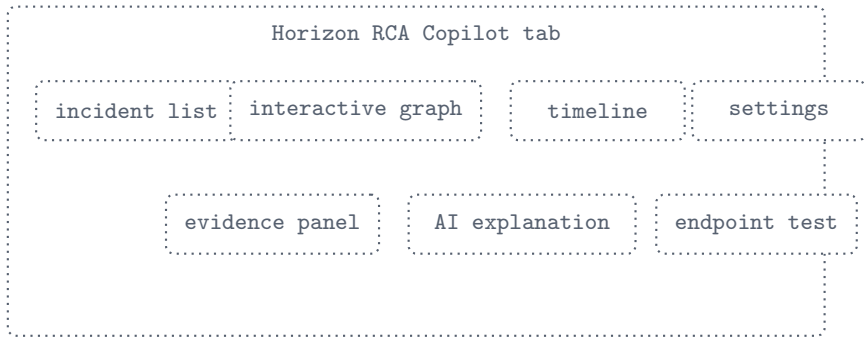
Grounded RCA Generation Flow



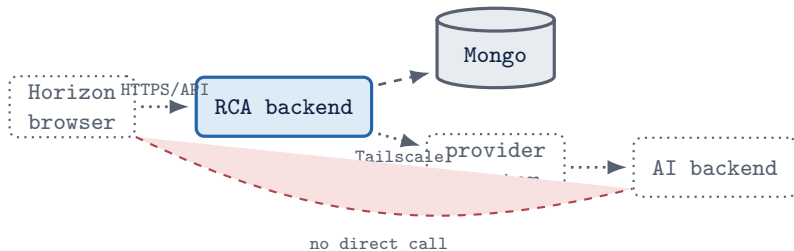
Unsupported Claim Gate



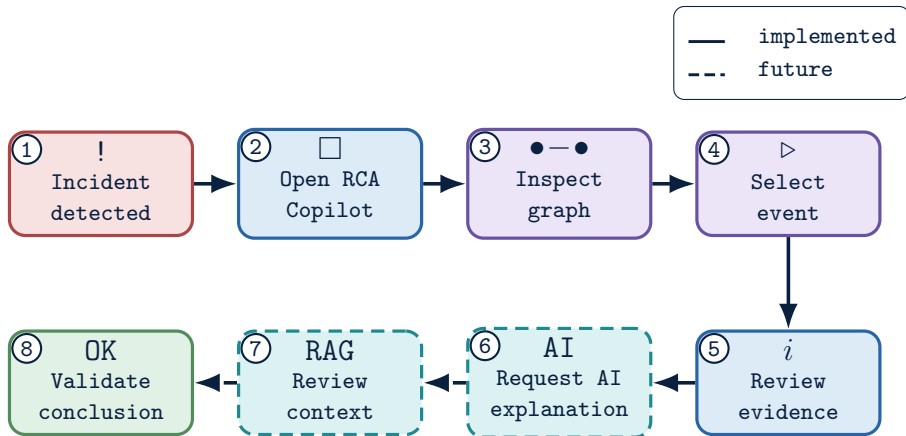
Future Horizon RCA Copilot Tab



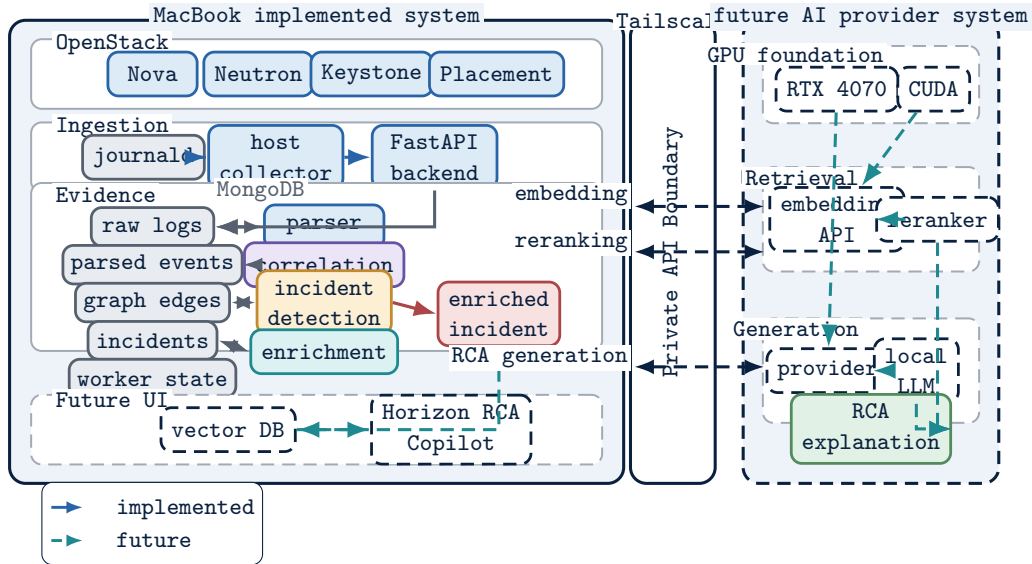
Browser-to-Backend Flow



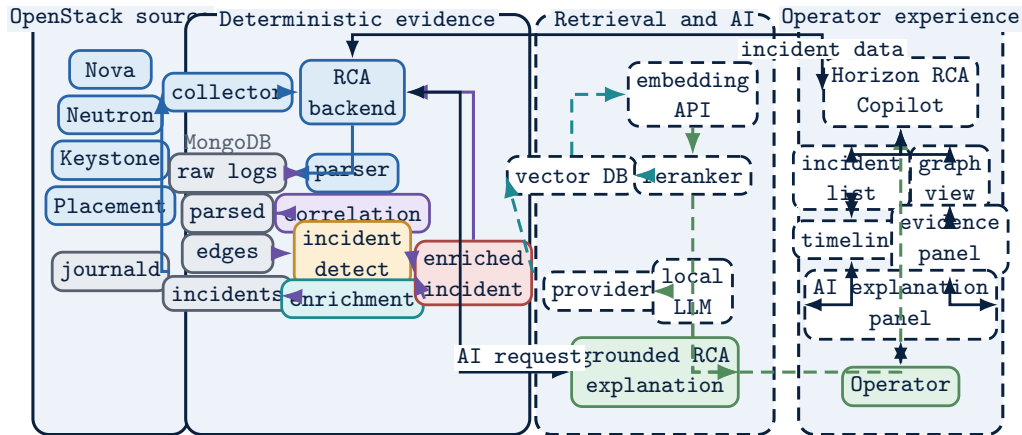
Operator Storyboard



Overall Architecture | Two-Machine RCA Copilot

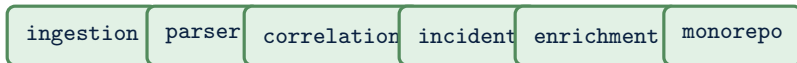


Full Final Architecture



Roadmap

Completed



Future



Evidence foundation first; local AI and Horizon integration after reliable artifacts exist.

Final Takeaway

OpenStack RCA Copilot turns
scattered service logs into a
traceable incident evidence graph.

graph first

evidence first

AI explains

The AI layer is valuable because the evidence layer is explicit, bounded, and auditable.